

Software	Version	Origin	Function within <i>miRFlow</i>
NumPy	1.26.0	NumPy Developers	Numerical computation, array processing, event filtering, threshold calculations, interpolation, and diagnostic ratio calculations
Pandas	2.0.3	Pandas Developers	Data manipulation, structured table generation, patient summary creation, and Excel-ready dataframe handling
SciPy	1.16.3	Open-source project	Kernel density estimation and peak/shoulder detection for MMB FSC-window identification and earlier DG fluorescence gating logic
scikit-learn	1.3.0	Scikit-learn Developers	Gaussian Mixture Model implementation for fluorescence-state clustering and probabilistic cluster assignment
Matplotlib	3.8.0	Matplotlib Developers	Generation of 2D flow cytometry plots, 1D density plots, QC plots, collated overview figures, and cluster visualisations
fcsparser	0.2.5	Open-source project	Reading and parsing .fcs flow cytometry files into Python-readable dataframes
flowutils	1.0.3	Open-source project	Logicle transformation for flow cytometry fluorescence and FSC signal processing, with fallback transformation logic where required
openpyxl	3.1.5	Open-source project	Creation, formatting, colouring, and post-processing of Excel workbooks containing GMM summaries and diagnostic metrics